

MagicCode Super-Resolution Software Getting Started

Document Version: v2.0.0. Release Date: September 2, 2026.

1. Choose an integration path

Path	Library	Header	Use when
Simple spatial	libmagic_sr_enable.a / libmagic_enable_sr.lib	mc_enable.h	One GPU texture per frame
Session (spatial or temporal)	libmagic_sr.a / libmagic_sr.lib	mc_interface.h	Explicit I/O, CPU buffers, or temporal

2. Spatial Enable (quick start)

Link the Enable archive, include mc_enable.h, set a model path, then:

```
void* out = MC_Enable(input_texture, 2.0f); /* then MC_Disable */
```

- Scale [1.0, 8.0]; <=0 defaults to 2.0.
- Android GLES/Vulkan: call MC_Enable_SetInputSizeHint first.
- Windows Enable default is OpenGL. MAGIC_BACKEND_D3D11 is not an Enable backend.

3. Temporal session (v2.0.0)

Do not start temporal from Enable. Link libmagic_sr, include mc_interface.h, set alg_mode to TEMPORAL_SPEED_MODE or TEMPORAL_BALANCED_MODE, fill gpu_context, then each frame fill magic_frame_t.image_in/image_out plus a temporal_frame_t (struct_size, depth, motion, jitter, frame_index, camera) and call MC_Process.

Temporal Super-Resolution

v2.0.0 adds temporal super-resolution through the session API: libmagic_sr.a (Windows: libmagic_sr.lib) + mc_interface.h + MC_Init / MC_Process / MC_Uninit. MC_Enable remains a single-texture spatial helper. It may accept a temporal alg_mode_e value by range check, but it cannot supply per-frame depth, motion, or jitter and must not be documented as a complete temporal entry.

Modes

Value	Enumerator	Role
0	SPATIAL_SPEED_MODE	Spatial, throughput first
1	SPATIAL_BALANCED_MODE	Spatial, quality/cost trade-off
2	TEMPORAL_SPEED_MODE	Temporal, throughput first

3	TEMPORAL_BALANCED_MODE	Temporal, canonical FSR-family path
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- Temporal scaler_factor is (1, 8]; 1.0 itself is rejected. Spatial remains [1.0, 8.0].
- TEMPORAL_SPEED_MODE on Metal, Vulkan, and OpenGLS currently uses a lightweight Convert → Activate → Upscale pipeline. Direct3D 11 and desktop OpenGL 4.3 Speed use the FSR-family temporal path (same family as Balanced, not a separate customer API).
- TEMPORAL_BALANCED_MODE uses the canonical FSR-family path on every GPU temporal backend. Desktop Balanced edge-resolve is supported on Windows Vulkan, Windows Direct3D 11, desktop OpenGL 4.3, and Apple Metal. Android Vulkan and Android OpenGLS do not run edge-resolve. If enable_sharpening is on, RCAS runs after edge-resolve.
- CPU backends (X86 / NEON) cannot run temporal modes.
- The API accepts (1, 8]. Exact 2× has optimized kernels. Not every scale or device is claimed as device-run.
- Internal pipeline names are implementation detail, not public configuration.

Minimum per-frame inputs

- magic_frame_t.image_in / image_out: caller-owned color and output GPU resources.
- magic_frame_t.frame: non-NULL temporal_frame_t* with struct_size = sizeof(temporal_frame_t).
- temporal_frame_t.depth and motion at input resolution.
- jitter_offset_x/y in input pixels, top-left, +X right, +Y down.
- frame_index / reset_history; camera_near / camera_far / camera_fov_y (radians).
- gpu_context at MC_Init: Vulkan requires physical_device + device; D3D11 requires device; Metal device may be NULL; GL/GLES use the current context (library never makes current).
- Vulkan temporal requires magic_frame_t.command_buffer to be a recording VkCommandBuffer. Metal command_buffer is optional. D3D11 / GL / GLES leave it NULL.
- Create-stage depth_reversed / depth_infinite / hdr_color on input_param_t (all-zero = conventional-Z, finite far, LDR). Immutable for the handle lifetime.

Motion, depth, and masks

- Motion is current → previous. The library never negates MVs.
- motion_vector_scale_x/y is the only MV numeric conversion. (0,0) defaults to (input_width, input_height). Mixed zero is rejected.
- Depth is GPU device depth in [0, 1]. Linear view-space depth is not accepted.
- reactive / transparency are optional. Explicit caller textures win per channel.
- When omitted, the library derives an approximate mask. That is not a semantic transparency mask.
- Speed on Metal/Vulkan/GLES may derive both missing reactive and an approximate transparency. FSR-family paths (Balanced on all backends; Speed on D3D11/desktop GL) derive missing reactive only and bind an internal zero transparency texture.
- Both temporal modes enable depth+MV history rejection by default.
- Diagnostic only: MAGICSR_TAAU_AUTO_MASK=0 and MAGICSR_TAAU_HIST_REJECT=0 can turn those defaults off. They are not a customer configuration path.

4. Models

GPU Enable bins are magic_<backend>_speed_gpu_params.bin and magic_<backend>_balanced_gpu_params.bin. Use tools/setup_models.sh (copies Speed + Balanced GPU bins). Call SetModelDir / SetModelPath in production apps.

5. Next documents

- User Guide: doc/User_Guide.md and doc/用户使用说明书.md
- API Reference, Integration Guide, FAQ, Migration Guide, Known Issues in this folder